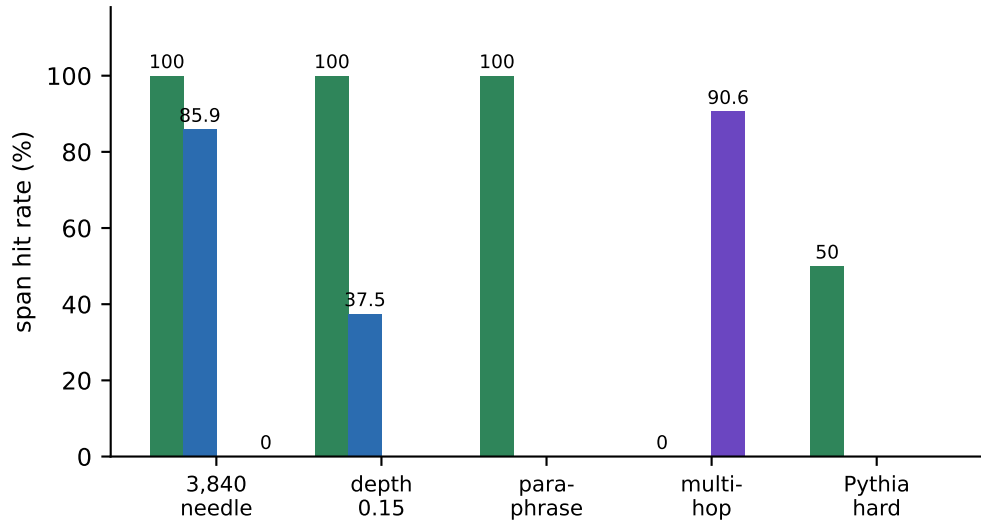


(a) Span detectors by substrate

lexical (training-free) L8 residual probe (labeled dev)
retrieval-head attention needle-affine experts



(b) Label-free pipeline recovery

